

created by the diode onto the media. The lens is connected with springs that allow small changes to be made to accommodate for small differences in the disc surface. The laser is reflected by the reflective layer, but based on the presence of a Land or Pit (or transparent or opaque area) on the media, the laser is reflected back into the sensor accurately, or is diffused. This difference in intensity of the reflected light is interpreted as binary digits (0s or 1s).

2.2.25 SAO

The Session at Once mode of writing, also called Multisession writing, writes data to the medium in the form of sessions, complete with the Lead-in and Lead-out. This allows the data on an incompletely-written disc to be read.

2.2.26 Sectors

Tracks on a optical media are subdivided into sectors, which are the smallest storage unit of the medium. In CD ROMs, a sector contains 2352 bytes, while in a DVD, the sector is 2048 bytes long.

2.2.27 Session

A session refers to the basic unit of data written to a disc. A session consists of a Lead-in, the actual data, and a Lead-out.

2.2.28 Speed notation

The original “X” that is used to denote reading and writing speeds in optical drives of all types refers to the speed of the first drive of that type. So a 52X CD drive can read or write data 52 times faster than the original CD drive, which means at about 150 x 52, or 7800, Kilobytes per second. The notation seen on optical drives refers to the speeds during different operations. For example, a CD-Writer with the 52X24X40X notation can read CDs at 52X, erase CDs at 24X, and write CDs at 40X.

2.2.29 Stamping

Optical discs are manufactured in bulk by first creating a die that contains the pattern of pits and lands that represent the data. This pattern is then transferred to the disc by “stamping” it onto the die.